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**T11 - 05**

### **Assignment No. 6**

**AIM** - Implement student registration form with validation using angularjs.

**LO3** – Implement Single page application using AngularJS

#### **Theory:**

A **Single Page Application (SPA)** is a web application that loads a single HTML page and dynamically updates the content as the user interacts with the app, without reloading the entire page. **AngularJS** is well-suited for building SPAs due to its powerful features like **two-way data binding**, **routing**, and **dependency injection**. Using the ngRoute module or ui-router, developers can define different views (or templates) and link them to specific controllers, enabling dynamic navigation between pages like home, about, and contact—while staying on the same HTML file.

To implement a SPA in AngularJS, the developer first defines routes using the \$routeProvider service inside the app's configuration block. Each route points to a template and a controller, which handles the logic for that particular view. When the user clicks a link or performs an action that triggers a route change, AngularJS dynamically loads the relevant view without a full-page reload. This provides a faster, smoother user experience similar to desktop applications. SPAs built with AngularJS are ideal for creating interactive and responsive web apps such as dashboards, chat apps, or e-commerce platforms.

#### **PROGRAM:**

index.html

<!DOCTYPE html>

```
<html lang="en" ng-app="studentApp">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Student Registration Form</title>

  <!-- Bootstrap CSS -->

  <link
href="https://stackpath.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css "
rel="stylesheet">

  <script
src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></scrip
t>

  <link rel="stylesheet" href="style.css">

</head>

<body ng-controller="StudentController">

  <div class="container mt-5">

    <h2 class="text-center">Student Registration Form</h2>

    <!-- Success Message -->

    <div class="alert alert-success" ng-if="formSubmitted">

      <strong>Success!</strong> Your form has been submitted successfully.

      <pre>{{ student | json }}</pre>

    </div>

    <!-- Form -->
```

```
<form name="registrationForm" ng-submit="submitForm()" novalidate>

  <!-- Full Name -->

  <div class="form-group">

    <label for="fullName">Full Name:</label>

    <input type="text" id="fullName" name="fullName"
      class="formcontrol" ng-model="student.fullName" ng-pattern="/^[A-Za-
z\s]+$/" required>

    <small class="form-text text-danger"
      ngshow="registrationForm.fullName.$touched &&
      registrationForm.fullName.$invalid">Full Name is required and must contain only
      letters.</small>

  </div>

  <!-- Email -->

  <div class="form-group">

    <label for="email">Email:</label>

    <input type="email" id="email" name="email" class="form-control"
      ngmodel="student.email" required>

    <small class="form-text text-danger"
      ngshow="registrationForm.email.$touched &&
      registrationForm.email.$invalid">Valid email is required.</small>

  </div>

  <!-- Phone Number -->

  <div class="form-group">

    <label for="phone">Phone Number:</label>
```

```
<input type="text" id="phone" name="phone" class="form-control" ng-model="student.phone" ng-pattern="/^[0-9]{10}$/" required>
```

```
<small class="form-text text-danger"
ngshow="registrationForm.phone.$touched &&
registrationForm.phone.$invalid">Phone number must be 10 digits.</small>
```

```
</div>
```

```
<!-- Date of Birth -->
```

```
<div class="form-group">
```

```
<label for="dob">Date of Birth:</label>
```

```
<input type="date" id="dob" name="dob" class="form-control" ng-model="student.dob" ng-change="calculateAge()" required>
```

```
<small class="form-text text-danger"
ngshow="registrationForm.dob.$touched && registrationForm.dob.$invalid">Date
of Birth is required.</small>
```

```
<small class="form-text text-danger"
ngshow="!registrationForm.dob.$invalid && student.age < 18">Student must be at
least 18 years old.</small>
```

```
</div>
```

```
<!-- Gender -->
```

```
<div class="form-group">
```

```
<label>Gender:</label><br>
```

```
<input type="radio" ng-model="student.gender" value="Male" required>
```

Male

```
<input type="radio" ng-model="student.gender" value="Female" required>
```

Female

```
      <small class="form-text text-danger"
ngshow="registrationForm.gender.$touched &&
registrationForm.gender.$invalid">Gender is required.</small>
```

```
</div>
```

```
<!-- Password -->
```

```
<div class="form-group">
```

```
  <label for="password">Password:</label>
```

```
  <input type="password" id="password" name="password"
class="formcontrol" ng-model="student.password" required>
```

```
    <small class="form-text text-danger"
ngshow="registrationForm.password.$touched &&
registrationForm.password.$invalid">Password is required.</small>
```

```
</div>
```

```
<!-- Confirm Password -->
```

```
<div class="form-group">
```

```
  <label for="confirmPassword">Confirm Password:</label>
```

```
  <input type="password" id="confirmPassword" name="confirmPassword"
class="form-control" ng-model="student.confirmPassword"
ngrequired="student.password" ng-pattern="student.password" required>
```

```
    <small class="form-text text-danger" ng-
show="registrationForm.confirmPassword.$touched &&
registrationForm.confirmPassword.$invalid">Passwords must match.</small>
```

```
</div>
```

```
<!-- Submit Button -->
```

```
        <button type="submit" class="btn btn-primary"
ngdisabled="registrationForm.$invalid">Register</button>
```

```
    </form>
```

```
</div>
```

```
<!-- Bootstrap JS & jQuery (Needed for Bootstrap 4's JS components) -->
```

```
<script src="https://code.jquery.com/jquery-3.5.1.slim.min.js"></script>
```

```
<script
src="https://cdn.jsdelivr.net/npm/@popperjs/core@2.5.2/dist/umd/popper.min.j
s"></script>
```

```
<script
src="https://stackpath.bootstrapcdn.com/bootstrap/4.5.2/js/bootstrap.min.js"></
script>
```

```
<script src="app.js"></script>
```

```
</body>
```

```
</html>
```

app.js

```
// app.js var app = angular.module('studentApp', []);
```

```
app.controller('StudentController', ['$scope', function ($scope) {
```

```
    // Initialize the student object
```

```
    $scope.student = {};
```

```
    $scope.formSubmitted = false;
```

```
    $scope.student.age = null;
```

```

// Function to calculate age based on the Date of Birth

$scope.calculateAge = function () {    if
($scope.student.dob) {        var dob = new
Date($scope.student.dob);        var today = new Date();        var
age = today.getFullYear() - dob.getFullYear();        var m =
today.getMonth() - dob.getMonth();

        if (m < 0 || (m === 0 && today.getDate() < dob.getDate())) {            age--;
        }

        $scope.student.age = age;
    }
};

```

```

// Function to handle form submission

$scope.submitForm = function () {    if
($scope.registrationForm.$valid) {
        $scope.formSubmitted = true;
    }
};

});

```

```
style.css body { font-family:
Arial, sans-serif; padding:
20px;
} label { display: block;
margin-bottom: 5px;
}
input { margin-bottom:
10px; padding: 8px;
width: 100%; max-width:
300px;
}
button { padding: 10px 15px;
background-color: #4CAF50; color:
white; border: none; cursor:
pointer;
}
button:disabled { backgroundcolor:
#ccc;
}
span { color: red; font-size:
12px;
}
```

```
.form-container {  maxwidth:
400px;  margin: 0 auto;

}

h2 {  text-align:
center;

}

.error {  color: red;  font-
size: 12px;

}
```

## OUTPUT:

### Student Registration Form

Full Name:

Email:

Phone Number:

Date of Birth:

Gender:

☐ Male ☐ Female

Password:

Confirm Password:

## Student Registration Form

Full Name:

Full Name is required and must contain only letters.

Email:

Valid email is required.

Phone Number:

Phone number must be 10 digits.

Date of Birth:

 

Date of Birth is required.

Gender:

☒

Male

☐

Female

Password:

Password is required.

Confirm Password:

---